



D2.1 GDDS architecture diagram

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D2.1 GDDS architecture diagram

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SUMMARY

The SAGE project team has agreed on the basis for a core architecture of the GDDS. This core architecture is the first step in the architecture definition of the project that is the responsibility of Task 2.1. This deliverable is not a report but a schema of the SAGE core architecture. However, to help the reader to interpret the diagram, this document was created to accompany the diagram and provide some context and a detailed explanation on how to interpret the diagram.



Document history

Version	Version date	Version notes
0.7	27.08.2025	First version of the architecture description.
0.8	30.08.2025	Document presented in WP2 and reviews received.
1.0	31.08.2025	Reviews addressed and formatting finalized.

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Acronyms

DS	Data Space
EA	Enterprise Architect
EMF	Eclipse Modelling Framework
FACTS	Federated, Agile, Collaborative and Trusted System
FDS	Forest Data Space
GDDS	Green Deal Data Space
GUIs	Graphical User Interfaces
IDSA	International Data Space Association
PNG	Portable Network Graphics
SMEs	Small and Medium-sized Enterprises
SoS	System of Systems
STAC	SpatioTemporal Asset Catalog
UML	Unified Modelling Language
XMI	XML Metadata Interchange



Executive Summary

This deliverable provides the schema of the SAGE core architecture for the Green Deal Data Space (GDDS). The core architecture is the first step towards the IT architecture definition for the GDDS created in the project. This work was done under the Task 2.1. While some informal interviews were conducted with the use cases, a formal requirements study will be conducted in Task 2.3 that will be useful to perfect the current diagram in the near future. While the deliverable is not classified as a report, this short document provides the schema to help the reader interpret the diagram and understand the concept.

The architecture is based on the following basic design principles: iterative, reusability, inclusivity, fairness, configurability, extensibility, modularity applicability and scalability. This translates in the following practical principles: Entry Barrier as low as possible, System of Systems, Balance between Standardization and Mediation, Data as entry point, Loose-coupling, Interoperability/Security Orthogonality, Ideally use open-source solutions, Serve the requirements coming for the use cases in SAGE first, and Use implementations that are available from project partners.

The core architecture is divided in 4 layers: consumer (representing some of the consumer actors and some user interfaces), control plane (contains the “data space connectors”, and it is focused on metadata, catalogues and authentication), data plane (data assets, vocabularies, and data models), backend (data/metadata processing necessary to harmonize, semantically annotate, or any other kind of trusted processing done over the data).

Most of the data assets are stored outside of the GDDS by their providers and they are processed outside the GDDS. Consequently, the proposed core architecture sets a GDDS that is neither a storage facility, nor a processing computing resource. Instead, the GDDS relays on external existing data and metadata and provides access through dataspace technologies and standards. However, some trusted processing will be provided that will generate some data derivatives that are going to be stored in the GDDS.

The core diagram includes the Data Space Connectors as the main component to enable trust in the GDDS, following the International Data Space Association and the Data Space Protocol. However, the project recognizes at least another alternative called FACTS. In FACTS, data assets are trustable because they are immutable and linked to their metadata and it uses credentials in a wallet to validate data transactions (eliminating the need for the contract negotiation). For the moment, this alternative is not part of the core architecture, but it could be introduced in the future.

This is the first step towards the description of the final architecture of the SAGE project to guide the deployment of the GDDS. The next evolution of this architecture will be provided by Task 2.3 no later than month 18 in the project. In addition, customized versions of the architecture will be developed to demonstrate that each case can be implemented with the current core architecture. Finally, Task 2.4 will monitor the evolution of the architecture and deliver a 3rd version in month 27 and others afterwards to month 32



1 Purpose of the deliverable

The main objective of WP2 of SAGE project is to design the GDDS integrated technical infrastructure, ensuring it supports selected applications and use cases across focus areas of biodiversity, zero pollution, climate change, and circular economy.

The purpose of this deliverable is to present the first GDDS core architecture diagram designed under Task 2.1 in the first 6 months of the project, the first step towards the description of the final architecture of the SAGE project to guide the deployment of the GDDS.

In addition to the diagram, the deliverable provides a description of the methodology followed for the design of the core architecture diagram (Section 2). This includes the description of high-level and architecture-specific principles of the GDDS, the analysis of the project's 10 use cases with examples of initial identified dataflows (described in Annex A), the extraction of storylines (presented in Annex B) derived from these dataflows, and the selection of the component UML (Unified Modelling Language) diagram. The tool used to elaborate the diagram is Enterprise Architect (EA, by Sparks Systems). Section 3 introduces the GDDS architecture itself. It details the separation into layers, the proposed architecture diagram, and the description of its main components. Both internal and external components to the GDDS are included, along with their relationships to other data spaces and actors. An alternative architectural option called FACTS is also proposed to provide flexibility and a broader perspective.



2 Methodology

2.1 Design principles

In the design of the GDDS core architecture; we have considered a set of principles that has guided our decisions. These principles can be classified into high level principles and architectural principles and are based on the GREAT design principles (Santoro and Mazzeti, 2024).

2.1.1 High level principles

We could consider that the provision of the necessary means that will make data space work is not the only high-level requirement. It concerns the ‘what’ dimension of the GDDS. We also need to consider the ‘how’ dimension which is related to the recognized high-level values. They can be expressed through a set of general principles acting as requirements and constraints of the GDDS. In a preliminary phase, we identified the following set of basic principles:

- Iterative:
 - The GDDS architecture will be gradually perfected and completed in phases. Each phase will be followed by an implementation that will generate lessons and new needs. It is necessary to adopt a process where standards and common technologies will be adopted. In this process it is also necessary to split the workload between providers and consumers. SAGE must manage its architecture in an evolving environment where new solutions can emerge. If we design for multiple implementations or a particular functionality, we are likely well prepared to also deal with rapidly developing new frameworks & components.
- Reusability:
 - Reuse existing approaches and solutions as much as possible.
- Inclusivity:
 - The data that should be part of the GDDS is highly heterogeneous in terms of supported metadata content and formats, data encoding, coordinate reference systems, ontologies, etc. Some of this heterogeneity is required by the specific communities that generate and use those data. Since the driving benefit of a data space is to share all the valuable datasets, data should not be excluded from the GDDS due only to their diversity (if they do not compromise the overall level of service of the GDDS).
 - Many data sources targeted to be part of the GDDS are already part of other systems of systems or ecosystems with their own mandate and governance – e.g., European research infrastructures, Copernicus Services, European Space Agency ground segments, public administration systems including INSPIRE. It is necessary to respect such autonomy without imposing, de-jure or de-facto, the exclusive participation in the GDDS. This is strictly related to the autonomy vs. belonging conflict that will affect any data system. In a Common European Data Space, belonging should be encouraged through soft means mostly based on the overall value of the data space.
- Fairness:
 - Data that should be part of the GDDS will come from all sorts of organizations, such as companies, SMEs, public administrations, research and academic organizations, intergovernmental institutions, data intermediaries, citizens and data altruism



organizations. A data space should be the common ground where collaboration and competition take place for the benefit of all but, overall, for the GDDS to serve data for generating knowledge. Therefore, no privileged access, beyond what the data providers themselves require, should be granted to anyone at the risk of changing the fairness of the data space.

- Use federated (decentralised) technical services that avoid a single point of failure (making the architecture resilient). The implementation could be centralized at the beginning but with the possibility of extending to federated/distributed service provision. Centralized decision making at a governance level would be supported (so that there is consistency).
- Configurability:
 - For each component, protocols/formats are specified. The actual implementation(s) only need to comply with these.
 - Focus on protocols and formats that are already implemented by components available today or in the short term.
- Extensibility and modularity:
 - In addition to core components, the architecture includes provisions for extensions, some of which are preconfigured.
 - No single point of failure. We should avoid the use of centralized components and allow it only to accelerate implementation and reduce the effort for anyone else.
- Applicability and scalability:
 - The architecture should adhere to practical application scenarios. The design must allow for use in "real" production or design environments, going beyond the mere demonstration required by the European Commission for project purposes. For this reason, economic, legal and operational sustainability must be considered.

2.1.2 Architecture principles

In addition to the above basic principles, we identified the following set of architectural design principles for the GDDS:

- Entry Barrier as low as possible: the GDDS allows a low entry barrier for both data providers and data consumers, including non-geospatial experts.
- System of Systems: the GDDS is designed as a System of Systems (SoS) to interconnect many independent, autonomous systems, frequently of large dimensions, to satisfy a global goal while keeping them autonomous.
- Balance between Standardization and Mediation: the GDDS will rely on interoperability standards, developed at community level, complementing it with mediation/brokering to enable cross-domain interoperability.
- Data as entry point: the GDDS focuses on the sharing and use of data, independently of how the data is generated (e.g., off-line, on-the-fly, etc.).
- Loose-coupling: the GDDS DE is enabled by a set of APIs which can be used by data consumers to leverage (and enrich) the GDDS resources. It will also allow for a modular design that enables iterative evolution.



- Interoperability/Security Orthogonality: the GDDS security architecture is orthogonal to the GDDS interoperability architecture (i.e., any change in one of them should not affect the other one). We will ensure security and privacy by design
- Ideally use open-source solutions (licensed products are not excluded if they are fit for purpose and reasonably priced in our context)
- Serve the requirements coming for the use cases in SAGE first. Support user groups. What happens if the “reuse” component does not meet completely the requirements?
- Use implementations that are available from project partners. If no suitable implementations are provided from within the project, we can use external open-source components or alternatives.

2.2 Use case analysis

The design of the diagram of GDDS core architecture has been guided by a dedicated use case analysis. Between the end of May and mid-July 2025, one-hour online interviews were held with each of the ten use cases of the SAGE project. The aim of these interviews (directly organized by WP2) was to identify and obtain a clear overview of each use case’s needs and requirements, and to gain a better understanding of their activities. The main outcome was the identification of at least one dataflow per use case. The dataflows identified during the interviews are presented in [Appendix A](#). Please consider this a preliminary material.

DISCLAIMER: The WP6 has organized a more structured mechanism to capture the use case information in written form called trial handbook. As the trial handbooks become populated, the information in them will become more complete and more precise than the one you can find in the [Appendix A](#).

Based on the insights gained from the interviews, which enabled a detailed understanding of the activities and the mapping of dataflows for each use case, a workshop was organised at the end of July with all use case partners. During this workshop, the initial GDDS core architecture diagram was presented, designed to integrate and accommodate the identified dataflows. The objective was to validate the alignment of the architecture with the specific needs of each use case, ensure that no functional or data needs were overlooked, and collect feedback to further improve the core architecture diagram.

Finally, to ensure that our understanding of the use case activities is aligned with the core architecture diagram and that the latter adequately accommodates the identified dataflows, a comparison was undertaken between our understanding of the use case activities and dataflows and the content of the *Trial Handbook*, the first chapters of which are partially included in deliverable *D6.1 SAGE Use case methodology and monitoring* (J. Sepúlveda et al.).

In conclusion, the dedicated online interviews with each use case, the workshop involving partners of the 10 use cases, and the comparison with deliverable D6.1 together provide robust validation that the GDDS core architecture diagram both accommodates and aligns with the technical and practical needs of all use cases.

2.3 Extraction of initial storylines





Based on the dataflows identified during the use case analysis and our understanding of the corresponding activities, a set of storylines has been extracted which are presented in [Appendix B](#). These storylines are formulated to capture the perspective of the different actors involved, translating their needs into clear and actionable narratives. To ensure consistency, they follow the standard sentence structure: “As a [role], I want [action] in order to [purpose].” This approach provides a structured way of linking user needs and functional requirements to the design of the GDDS core architecture, while reflecting the concrete needs of the use cases. In the next months, these storylines will be refined and perfected.

2.4 UML

UML (Unified Modelling Language) is a standardized, general-purpose visual modelling language used to describe, specify, design, and document software systems and business processes in a clear and structured way. It is process-independent and widely used by business analysts, developers, and architects to visualize both the structure and behaviour of complex systems. UML provides a set of graphical notation techniques, like diagrams, to model the static and dynamic aspects of a system, enabling teams to plan and communicate software architecture efficiently. It is maintained by the Object Management Group (OMG) and covers numerous application domains, including business processes and IT systems.

UML diagrams are typically divided into two main categories: structural (static) and behavioural (dynamic), with a total of 14 standard diagram types:

- Structural Diagrams
 - Class Diagram: Shows the classes, their attributes, operations, and relationships.
 - Object Diagram: Depicts instances of classes (objects) and their interrelations at a particular moment.
 - Component Diagram: Models the physical components/modules and their dependencies.
 - Composite Structure Diagram: Represents the internal structure of a class or component.
 - Deployment Diagram: Illustrates hardware nodes and software artifacts deployed on them.
 - Package Diagram: Groups related classes or other elements into packages.
 - Profile Diagram: Describes metamodel extensions for custom domains
- Behavioural Diagrams
 - Use Case Diagram: Visualizes actors (users or systems) and the use cases (interactions) with the system.
 - Sequence Diagram: Represents object interactions arranged in time sequence.
 - Communication Diagram: Emphasizes object interactions and their relationships.
 - Activity Diagram: Models workflows or business processes.
 - State Machine Diagram: Describes state changes triggered by events.
 - Interaction Overview Diagram: Provides a high-level summary of control flow within interactions.
 - Timing Diagram: Shows object behaviour over time, focusing on timing constraints.

UML's diverse set of diagrams and notations allows teams to capture, analyse, and refine nearly every aspect of a system, from its overall architecture to intricate state transitions and user interactions.



2.4.1 Component diagram

To elaborate the core architecture of SAGE, we have selected the components diagram. A component diagram in UML is a visual representation used to show the organization, interrelationships, and dependencies among the modular pieces ("components") of a software system, emphasizing how they interact via interfaces and connectors. Component diagrams help model the static structure of a system at a high level, illustrating how software modules or subsystems fit together to form the complete architecture. They are used during both design and implementation phases to communicate with stakeholders and guide development.

The key elements of a Component Diagram are:

- **Component:** The primary building block, represented as a rectangle with the «component» stereotype; it encapsulates a set of functionalities and can contain other components or subsystems.
- **Interface:** Circles ("lollipops") indicate provided interfaces, and half-circles ("sockets") denote required interfaces, defining how components communicate with each other.
- **Port:** Squares on the component boundary, representing specific interaction points exposed for communication.
- **Relationships:** Connections and dependencies between components and interfaces, shown by solid or dashed lines and arrows that visualize interactions, such as assembly connectors linking provided and required interfaces.
- **Artefacts and Nodes:** Less commonly, diagrams can include physical files («artifact») and execution environments («node») to clarify deployment or runtime structure.

Component diagrams have several benefits: They ensure modularity, reusability, and clear interface definitions; facilitate stakeholder communication and development planning; and support system scalability and maintainability by decoupling component implementations. A component diagram is essential in software architecture for modeling the system's structural organization and high-level dependencies, bridging the gap between requirements and implementation.

2.4.2 Elements from other UML diagram types.

A component diagram in UML can incorporate or borrow elements from other UML diagram types, such as interfaces, ports, and relationships that are also found in class or deployment diagrams. Component diagrams primarily focus on system architecture and the organization of software modules but share notation and concepts with other structural diagrams. Tools and standards allow embedding composite diagrams or referencing sub-components defined elsewhere, facilitating reuse and consistent architectural modelling. Component diagrams are flexible, allowing them to borrow graphical or conceptual elements from other UML diagrams to provide a comprehensive view of system architecture and its interaction points.

In the elaboration of the components diagram, it was necessary to bring elements from other UML diagrams. For example, we used horizontal "partitions" from the UML activity diagrams to separate the components in layers and the "actor" from the UML use case diagram.



In addition, when more than one instance of a component exists, the core diagram represents them all as a single “abstract” component; sometimes adding some instances as specializations of the “abstract” one.

2.4.3 Tool used for generating the UML diagram.

The component diagram has been elaborated in UML with Enterprise Architect (EA, by Sparks Systems¹) version 9 (2011). EA v9 uses a particular table model in Microsoft Access which has an EAP extension. With EA the model can be saved to PNG, EMF or exported to the standard XMI file format. Furthermore, the project can be imported to more recent versions of EA and can be converted to the new extension for projects QEAX file format (based on SQLite database).

¹ <https://sparxsystems.com/>



3 GDDS architecture

3.1 Layer separation

Due to the number of components and functionalities required by the use cases, we decided to separate the architecture into different horizontal layers:

- Consumer (client): This layer represents some of the actors that act as consumers to the GDDS (human or machine). It also contains the user interfaces that human actors require.
- Control Plane (middleware): This layer represents most of the functionality that is described in the concept of “data space connectors”. It is focused on metadata, catalogues and authentication. It contains the endpoints for the discovery and access protocols.
- Data Plane (providers): This layer contains mainly data assets, as well as other assets such as vocabularies and data models. Most of the data assets are stored outside of the GDDS by their providers but it is possible that the GDDS will store some processed assets.
- Backend (supporting processing): This layer contains the data/metadata processing necessary to harmonize, semantically annotate, or any other kind of trusted processing done over the data in the GDDS.

Apart from the horizontal layers, there is an implicit vertical separation of:

- external provider components (mainly for open data) to the left of the diagram,
- the SAGE GDDS components for the core of the GDDS in the centre, and
- other data spaces to the right of the diagram.

3.2 Architecture diagram proposal

This is the core architecture diagram (Figure 1) proposed as a starting point for the project implementation of the GDDS. All components in the GDDS are illustrated with green background. External components to the GDDS are illustrated with orange background (the EA default colour).

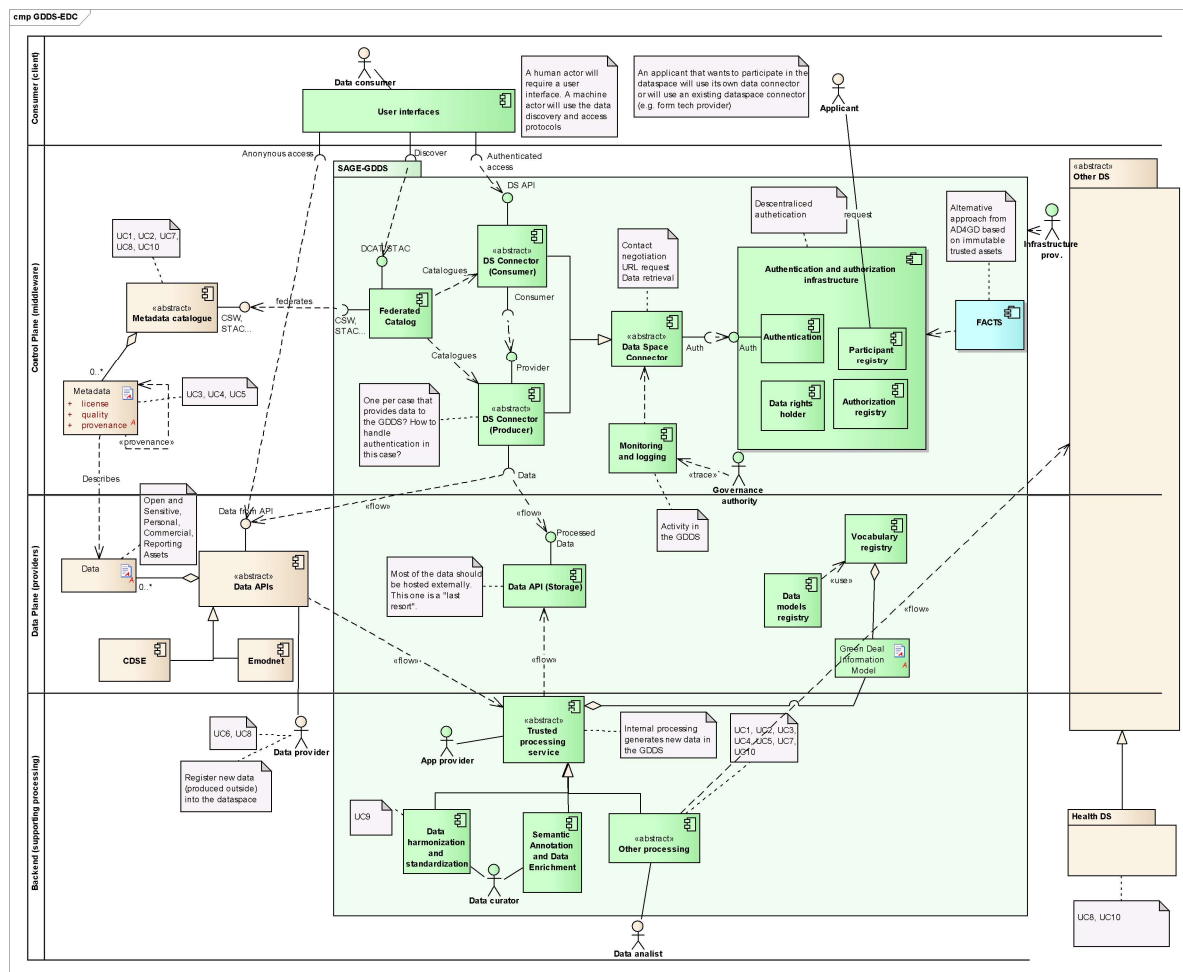


Figure 1: Core Architecture diagram

3.2.1 Basic Components

The centre of the diagram (and inside the SAGE-GDDS) is occupied by the Data Space Connectors that follow the Data Space Protocol initially specified by the IDSA and now transferred to the CEN-CLC/JTC 25 'Data management, Dataspaces, Cloud and Edge committee' (as a contribution to the European Trusted Data Framework) , with the contract negotiation, the URL request and the data retrieval (see Figure 2). This has the following components:

- Data Space Connector (C): Connector that the data consumer is connected and authenticated to. This connector has access to other connectors in the data space being able to do contract negotiation, URL request and data retrieval.
- Data Space Connector (P): Connector providing access to data assets. This connector is accessed by other connectors in the data space that will request the data managed by this connector.
- Federated Catalog: Federated catalogue enabling to discover datasets provided by all data space participants through data connectors.

NOTE: the distinction between a connector for a provider and a connector for a consumer is only conceptual. In general, the same connector instance should be able to support use by either a provider or a consumer or both.

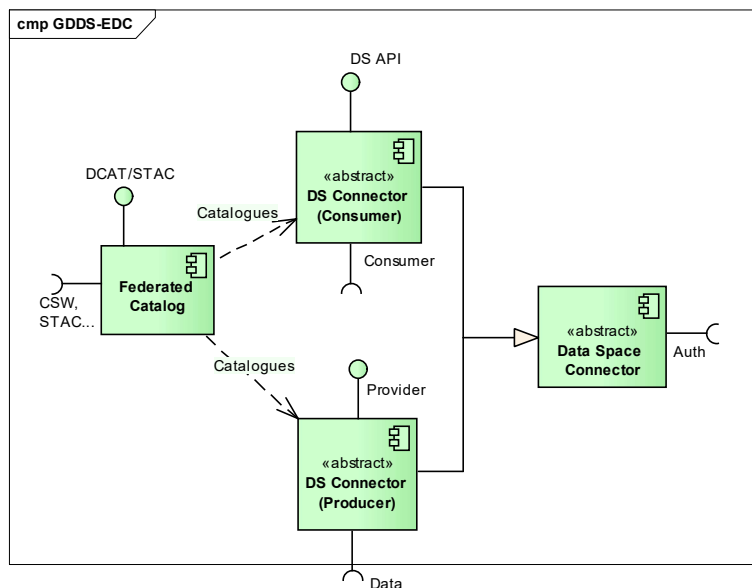


Figure 2: Dataspace Connectors (such as Eclipse Dataspace Connector)

The data connectors require an authentication process to have trust in the participants in the data space. The authentication allows for authorization of a specific user to do certain things as well as to negotiate the contracts. Commonly there are already good alternatives for authentication (such as Keycloak) and connector implementations rely on them instead of re-implementing them. Figure 3 shows the components related to authentication and authorization:

- Authentication: Gives the capacity to identify users as well as assign and verify tokens associated to users.
- Data rights holder: Provides the authorization capacity by storing the rights that the authenticated users have.
- Participant registry: Lists the participants in the GDDS.
- Authorization registry: Contains types of authorizations that the data rights holder will manage.

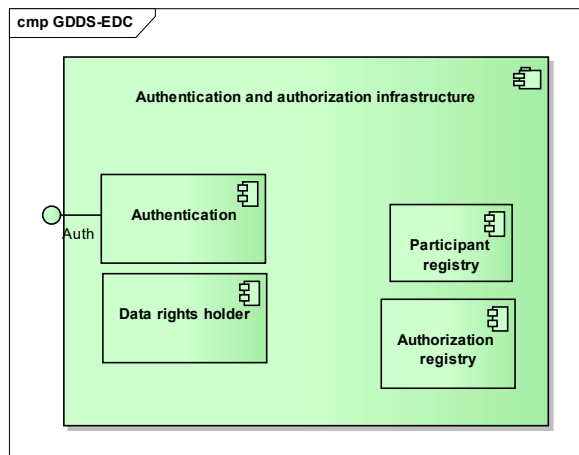


Figure 3: Authentication and authorization components.



Finally, there is another common building block in the Data Space that contributes to the trust: the Monitoring and logging that records all the activity of the data space connectors including the data transfer and how it was authorized.

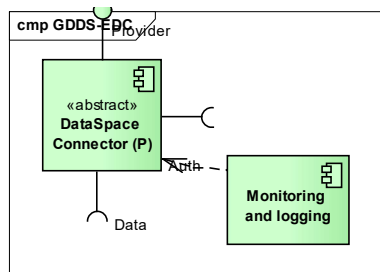


Figure 4: Monitoring and data logging components.

This completes the core building blocks of a data space.

3.2.2 External components

In SAGE we agreed that most of the data assets will be hosted in external services that will be integrated in the GDDS. Many assets will be part of the European open data policies, and some others will be contributed by companies as protected data (e.g. sensitive data, personal data, commercial data, reporting assets). These datasets are already made accessible by data services, each one implementing their restrictions. Their metadata have also been made available and discoverable. These datasets can be contributed to the data space via a data connector, and their metadata records can be replicated in the GDDS federated catalogue.

At this point, SAGE has decided that even if the data is open, it will be declared part of the GDDS and made accessible via the data connectors and the federated catalog. In this way the GDDS will be able to monitor their usage and record the access of trusted users. Of course, we will not prevent any other form of data discovery and access that can be in place by the external provider. All this is represented by the following components as shown in the Figure 5. A user can connect directly to the metadata and data APIs directly if he is accepted or use the GDDS to do so.

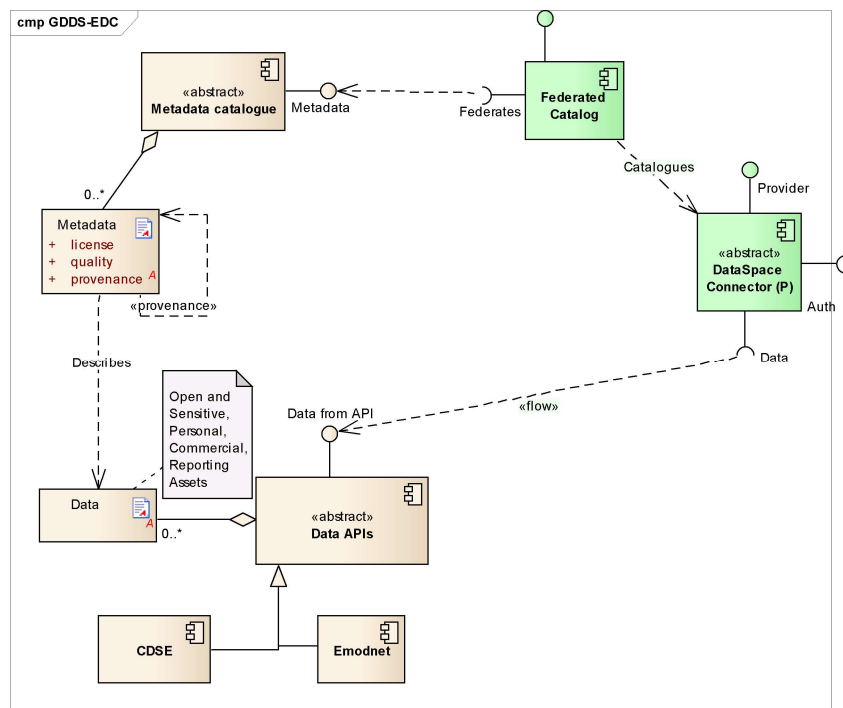


Figure 5: External metadata catalogues and data APIs that are connected and integrated in the GDDS.

3.2.3 Other GDDS components

In SAGE we agreed that we will push for a GDDS that will not have its own storage and most of the data processing will be done externally. However, we recognize from experience in other projects that some data processing can improve performance in the data space, such as a semantic enrichment of the datasets. This is a process that needs to be done on the original data that creates another distribution of the data that needs to be stored somewhere in the GDDS. In conclusion, while the GDDS will not be considered a data storage facility or data processing machinery, some data storage and data processing will be needed. In that respect, only trusted processing software will be executed by the GDDS. This is the list of "other" components that the GDDS core architecture proposes in addition to the basic components:

- Data API storage: Data API and storage service that will be used by the trusted processing services only.
- Trusted processing services (abstract): Processing services that have been validated and can be trusted in the sense that they do not tamper with the data.
- Data harmonization and standardization: Process that transforms the data format into a more transversal and interoperable one that most users of the GDDS can use.
- Semantic annotation and data enrichment: Process that transforms the data into a semantically tagged format that most users of the GDDS can understand.
- Green Deal Information Model: A neutral data model that contains the concepts that are needed to understand most of the GDDS relevant data.

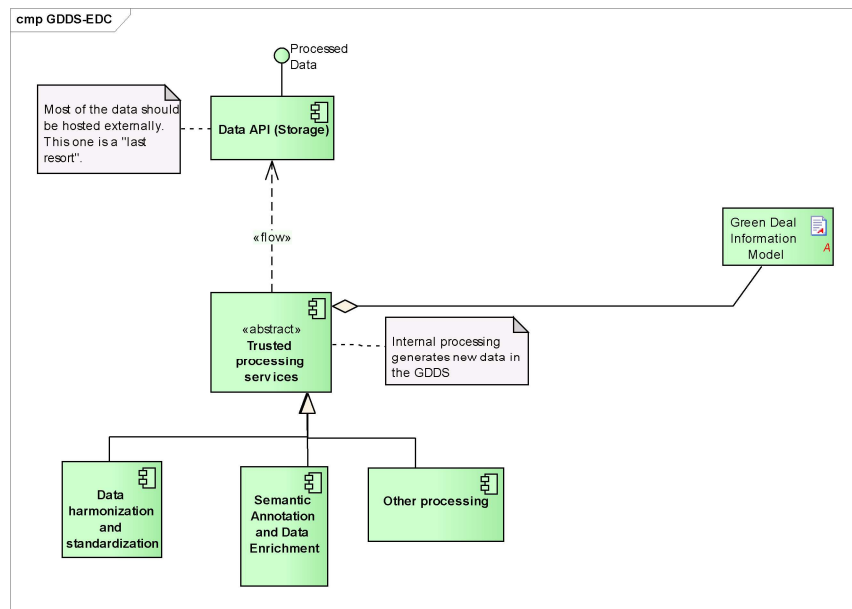


Figure 6: Other GDDS Components.

3.2.4 Other Data Spaces

Two SAGE use cases (UC8 Air Quality and Health Dataspace and UC10 Environmental hazards) have a broad set of requirements that the GDDS data will need to be combined with other DS and in particular to the Health DS. That is why other data spaces are represented next to the GDDS.

3.2.5 Actors

Several actors have been identified during the elaboration of this core architecture diagram. They are:

- Application provider
- Data analyst: An actor that brings a trusted processing service to the GDDS.
- Data consumer: An actor that wants to get access to some data in the GDDS or from external sources.
- Data curator: An actor that curates the data by means of executing data harmonization and data annotation trusted services
- Data provider: An actor that provides data to the GDDS through external data APIs.
- Infrastructure provider: An actor that manages the components of the GDDS.
- Governance authority: An actor that monitors and verifies the logs in the GDDS.

Actors will require user interfaces (GUIs) to access the different interoperable components. This is represented once in the Data consumer as seen in Figure 7.

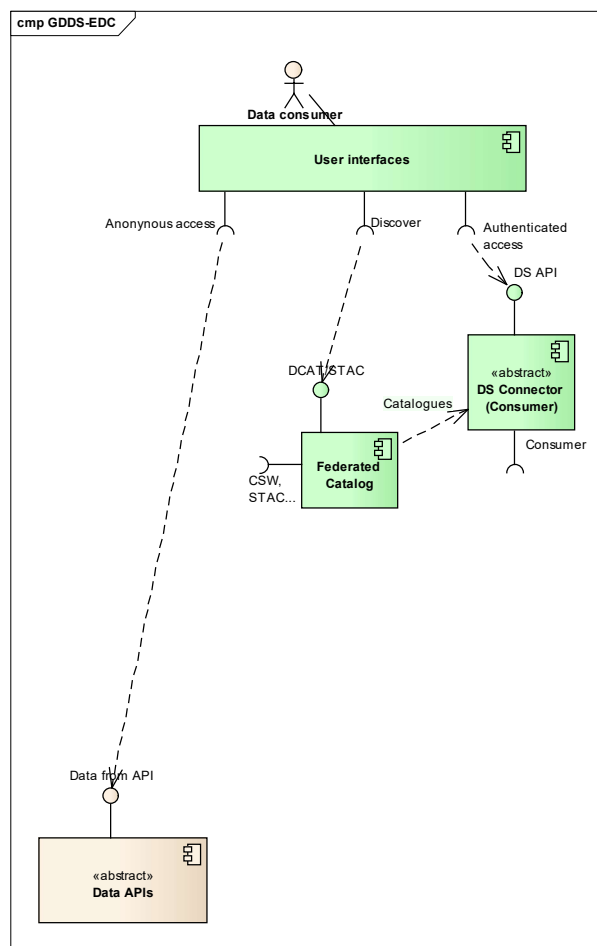


Figure 7: Need for user interfaces to access the data in the GDDS or externally.

3.3 Alternative architecture

The core diagram includes the Data Space Connectors as the main component to enable trust in the GDDS but recognizes other alternatives are possible. In the AD4GD project some experimentation was done on the FACTS approach (see Figure 8, Churchyard, 2024). In FACTS an immutable catalogue is used in combination of credentials in a wallet to validate data transactions eliminating the need for the contract negotiation. In FACTS, data assets are trustable because they are immutable and linked to their metadata. For the moment, this alternative is not considered but it could be introduced in the future.

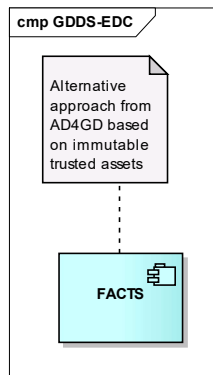


Figure 8: Alternative to the basic components in the GDDS.



4 Conclusions and next steps

The SAGE project has agreed on the basis for a core architecture. This core architecture is only the first step in the architecture definition of the project done by Task 2.1. Due to the time constraints, this core architecture was elaborated without a formal requirements process. However, consultations with the use cases were done and the needs of the cases were extracted in a less formal way. The project will now start a more formal process of requirements gathering in Task 2.3 and a new architecture will be released no later than month 18. In addition, customized versions of the architecture will be developed to demonstrate that each case can be implemented with the current core architecture.

This is the first step towards the description of the final architecture of the SAGE project for the deployment of the GDDS. Task 2.4 will monitor the evolution of the architecture and deliver a 3rd version in month 27 and others afterwards to month 32. See the evolution of the architecture in time illustrated in Figure 9.

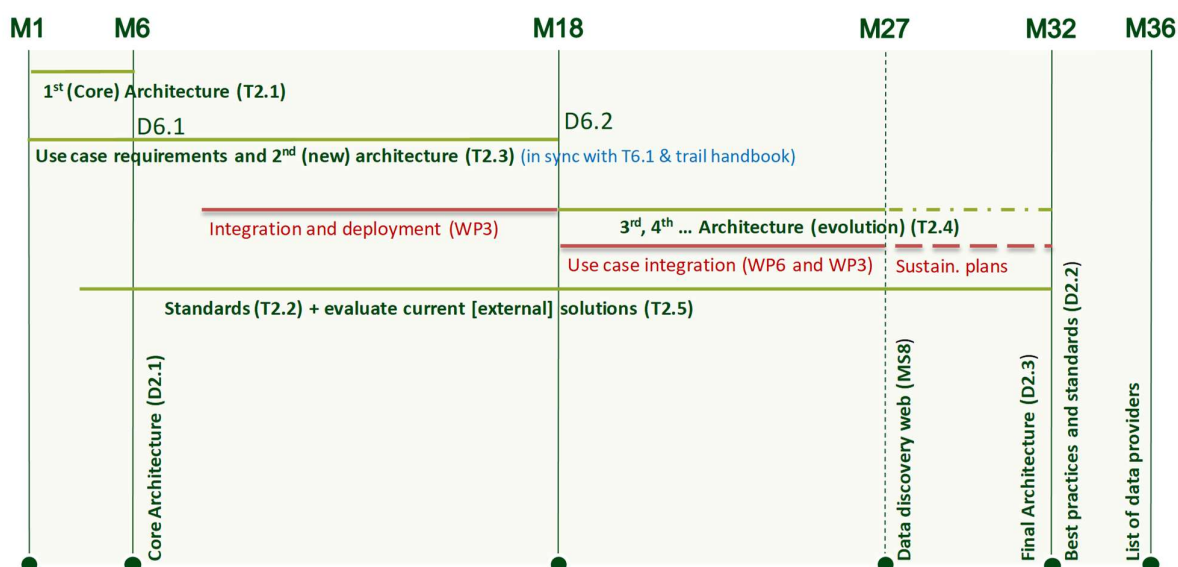


Figure 9: Evolution of the SAGE GDDS architecture in time.



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² On 7 July, CEN and CENELEC officially accepted the European Commission's Standardization Request. See: <https://www.cenelec.eu/news-events/news/2025/brief-news/2025-07-11-data-act-standardization-request/>



Appendix A: Use case interviews - identified dataflows

DISCLAIMER: SAGE has organized a more structured mechanism to capture the use case information in written form called trial handbooks. As the trial handbooks become populated, the information in them will become more complete and more precise than the one you can find in this section.

Interview UC1. Forest Dataspace (FDS) Cluster

Presenter: Thorsten Reitz (WE)

Date of Interview: May 20, 2025.

UC Overview:

This use case focuses on the integration of the existing German Forest Data Space (FDS) developed over four years, into the Green Deal Data Space. The aim is to improve forest management and forest transformation planning across Europe using harmonised, secure, and interoperable environmental and forestry data.

They have identified three key questions that forest practitioners need to answer:

- Which forest stands are to be prioritised for forest conversion?
- Which tree species or combination of tree species are suitable at a certain location to withstand future climatic requirements?
- How can forest conversion be achieved when considering biodiversity, profitability and climate function?

The FDS dealing with climate change addresses the actors listed below. All these organisations need to benefit from a data ecosystem. WE (wetransform) has identified what are the benefits to every single actor that is contributing to the FDS and they have identified the unit of value of each actor.

Actor	Unit of value
Data-holding organisations → GIS data providers.	Data pool
Multipliers → especially environmental authorities who want to make sure that there is a good foundation for sustainable decisions for everyone involved.	Impact
End users → private forest owners and forest practitioners, State forests, forest administrations.	Stand
Value-added services → seed suppliers, tree nurseries, inventory apps, UAV operator, machine rental company.	API calls

Geographical scope: For now, they process data from Germany, but they are planning to expand to other EU countries such as Finland, Switzerland, Spain, and Hungary.

Stakeholders involved: The main stakeholders involved in this use case are policy makers and funders, citizen and social societies, public administrations and NGOs, industry and SMEs and research and academic institutions.



General data aspects: It consists of datasets available on forest cover, health, biomass, disturbance, status, trends etc. The data is used to monitor forest resources, assess timber stocks, track changes in forest area, and understand forest health. Some examples of data types are raster maps, GeoTiff datasets, NFI-III API, and aggregated forest statistics at local and national level.

WE (wetransform) has defined a semantic definition of core FI data for local forest inventory making sure that this model is also compatible with larger scale data such as National Forest Inventories (NFI). They have already a distributed FDS architecture where a data trustee can be the entry point to the data space.

Data access: Part of the data they process is open, standardised and with low sensitivity, but most part of the data is not easily accessible. Specially the National Forest Inventory datasets are considered sensitive data by the owners. This data must be processed in a secure way otherwise nobody is going to give this data to others.

Data flows identified: 3 dataflows have been identified:

1. Synthetic training dataset creation with machine learning

- Input: sensitive forest inventory data (e.g., tree species locations) + satellite remote sensing data.
- Processing:
 - Synthetic labelled data is created to preserve privacy.
 - Combined with non-sensitive real data.
- Output: training datasets (synthetic + non-sensitive real data).
- Access: only synthetic outputs are shared with end users.

2. Forest Transformation Simulation

- Input: harmonised heterogeneous vector data (in 100x100m grid), soil and climate datasets.
- Processing: simulation of future forest estates/evolutions under different management and planting scenarios.
- Output: derived datasets used by forest managers for planning.
- Access: only derived data is shared (original sensitive data stays protected and only the original data owners will have access to it). Access through an API (not an OGC API). It's a simplified API adapted exactly to their needs.

3. Continuous Tree Monitoring

- Input: live time tracking of forest data (of the silvicultural part of the process - until the tree is ready to be harvested).
- Processing: process dataset that indicates loss or fatality.
- Output: continuous life inventory
- Access: it's sensitive data and must stay protected. Only the forest owners will have access to the data.
- Goal: Move from 10-year static inventories to live/real-time monitoring

Interview UC 2. Pollinator Monitoring Dataspace (PMDS)

Presenter: Marcin Plociennik (PSNC) and Zbigniew Koltowski (PSNC), Raul Palma (PSNC).

Date of Interview: May 20, 2025.

UC Overview:





The Pollinator Monitoring Dataspace (PMDS) aims to stimulate the transition towards more sustainable and pollinator-friendly agriculture by improving the understanding and identification of pollinator-friendly cultivars through a system for monitoring pollinator and plant diversity. The monitoring will be based on multiple combined data sources.

Geographical scope: The initial implementation will focus in 10 farms in Poland.

Stakeholders involved: Main stakeholders are farmer communities, agricultural businesses, EU pollinator initiatives, research institutes, governmental bodies (Ministry of Agriculture), the Institute of Plant Protection (IPP), Citizen Science projects.

General data aspects: Data integration and harmonization is necessary for this pilot due to the number of data sources. Currently, the monitoring is done manually but they want to develop digital tools for farmers to check on the field and evaluate the number of pollinators. They want to provide data support for AI models for pollinator detection.

The input for the Pollinator Data Space can be both data and services:

- Data: geographical data sources, information about crop varieties, taxonomic data, visual observations and acoustic data, soil quality maps.
- Data Services: data-based services (existing apps like iNaturalist, eBMS, etc), existing pollination optimisation services, Decision Support Systems in agriculture. In Poland they have the eDWIN (International Platform for Advice and Decision Support in Integrated Plant Protection in Poland).

Data access: still needs to be defined what data is public/sensible/private.

Data flows identified: this UC is still at a rather design stage. They have different potential applications. They are still in the process of identifying which datasets would make sense. The following are potential dataflows that this UC has:

1. Pollinator monitoring for farmers

- Input: pollinator monitoring data, crop types and field management.
- Processing: AI algorithm.
- Output: Recommendations based on pollinator presence or absence (e.g., when to rent hives, apply treatments)
- Access: farmers have access to the raw pollinator data of their fields as well as the output derived datasets and analysis.

2. Pollinator monitoring for policy makers

- Input: Aggregated data on pollinator health.
- Processing: ---
- Output: data/information about crops environmental health.
- Access: policy makers have access to these derived datasets for taking evidence-based and pollinator-friendly decisions.

To summarise, they still need to translate the proposal into more concrete small steps on:

- What are the data inputs?
- What is the processing?
- Where does it get stored?
- Who does it get sent to?



- Who might pay for it?

Interview UC3. Soil Circularity Dataspace (SCDS)

Presenter: Stefan de Graaf (sogelink) and Michel Grothe (sogelink).

Date of Interview: May 22, 2025.

UC Overview:

This use case focuses on enabling smarter, sustainable soil management by connecting soil suppliers (from excavation sites) with consumers (construction projects). The main goal is to make the soil flows in the Netherlands as circular and sustainable as possible.

This UC has developed a platform (<https://grondstromen.beta.geodan.nl/>) which is a multi-tenant digital marketplace allowing stakeholders to register excavation activities, assess soil quality, manage transport logistics, and identify reuse opportunities. The platform integrates public and private data sources and supports stakeholder collaboration in alignment with national and European sustainability goals.

Geographical scope: currently implemented in three regions (at the local level) in the Netherlands. They are planning to expand to other Dutch regions as well as at the national and European level.

Stakeholders involved: The main stakeholder involved is the public sector (governmental bodies) but the private sector will also be participant of this marketplace specially from the demand perspective.

General data aspects: need to integrate different types of data gathered from multiple sources.

Data flows identified: dataflows identified:

1. Soil excavation registration and analysis

- Input: polygon layer of excavation site and soil quality data.
- Processing: chemical analysis of soil at a particular site.
- Output: data on soil composition and soil quality. Map of soil type and contamination risk.
- Access: registered users only (contractors, municipalities) through the platform interface. APIs are used for automated processing.

2. Soil transport tracking and traceability

- Input: type and quantity of soil being moved at a certain location, time of soil transport.
- Processing: aggregated transport data and cross-check transport consistency (detect mismatches, illegal mixing).
- Output: soil transport records with full traceability chain.
- Access: only registered users can access the platform. In this case an environmental service has access to the data output. They are responsible of monitoring and enforcing environmental regulations regarding land use, soil quality and pollution control.

3. Circular soil use coordination

- Input: soil availability (type, quality, volume, timeline) from excavation records.
- Processing: match soil demand with available batches based on soil requirements (type, quality, volume, timeline) considering transport feasibility using multimodal routing services.
- Output: suggested matches between supply and demand and potential transport routes and timelines.
- Access: Registered users (municipalities, contractors)



Interview UC4. Construction Environment CO2 Hub & Building Twins (BECO2)

Presenter: Vesa Ilmarinen (Katalysti) and Antti Harjunpää (iSHARE) and Boróka Péter (iSHARE).

Date of Interview: May 27, 2025

UC Overview:

This use case focuses on real estate maintenance and building operations within the Finnish market, supported by a coalition of companies and organizations ([KiraHub](#) and [RAKLI](#)). The goal is to enable data sharing and reporting of [Scope 3 greenhouse gas \(GHG\)](#) (category 1 - Purchased goods and services, and category 5 - Waste generated in operations) emissions using a data space infrastructure, which supports semantic interoperability, secure data access, and compliance with reporting regulations (like the Greenhouse Gasses Protocol).

Geographical scope: Finland.

Stakeholders involved: real estate owners (RAKLI – association of Finnish property owners), service providers and material providers.

General data aspects: this UC has to understand the (minimal) requirements by the companies. The data they are dealing with are primary (it's not an average calculation) and undirect emissions from services and materials to buy from someone and the emissions when someone is using your product. Real estate companies are currently using Excel files (each company using different fields). Currently, moving towards harmonized APIs is an ongoing effort. There is also a strong need for semantic interoperability rather than just format standardization.

Data access: data may not always be public; access control is essential.

Data flows identified: the following dataflows have been identified:

- 1. Emission data from one service providers (logistics, materials and waste) to many real estate owners**
 - Input: primary CO₂ emission data from service providers (e.g., cleaning, general property maintenance, waste disposal, etc.) and metadata about the emission CO₂ data (accuracy, quality, how emission data was calculated, time periods and other metadata aspects).
 - Processing: CO₂ emission data calculations.
 - Output: reporting solutions focused on emission data or more generic reporting solutions to manage real estate portfolio data.
 - Access:
- 2. Carbon budget reporting flow**
 - Input: CO₂ generation from service providers.
 - Processing: emission budget computation.
 - Output: CO₂ emission reports.
 - Access: public administrations.

Specific needs: Rule book (how the ecosystem works), data catalogue (emission data available), data models (harmonised data models), authentication (identities and access management).



In some cases, the data of the product CO₂ emissions is public and in some other cases not. For example, when the companies report annually, the data can be public. There is also sensitive and non-public data such as more detailed information when they manage their operations.

Interview UC5. Textile Cluster in GDDS

Presenter: Traci Kinden (TEXroad)

Date of Interview: June 16, 2025.

UC Overview:

The Textile Cluster under Green Deal Data Space brings together public and private actors— Producer Responsibility Organizations (PROs), Digital Product Passport (DPP) providers, brands, recyclers, policymakers, and digital platforms—into a secure and connected ecosystem where textile data becomes interoperable, reusable, and a driver of effective circular action.

Geographical scope: public and private sector in European countries, with relevance across textile value chains.

Stakeholders involved: public and private actors—PROs, DPP providers, brands, post-consumer textile collectors and sorters, recyclers, digital platforms, and national level authorities.

Data flows identified: the following dataflows have been identified:

1. Value chain data harmonization and aggregation

- Input: [materials flows information]
- Processing: harmonizing textile value chain data within EPR monitoring systems
- Output: comparable company level reports and aggregated data sets
- Access: [data owners, data processors, public authorities]

2. Pull reverse supply chain data into the DPP

- Input: [recycled content information, certification evidence]
- Processing: integration into the DPP
- Output: [automated data upload for key data fields]
- Access: [data owners, data processors]

3. Push DPP data to reverse supply chain

- Input: product information stored in DPP (composition, product type, etc).
- Processing: Read and access DPP data to enable sorting, use DPP data for eco-modulation fee implementation
- Output: [relevant data for sorting, relevant data for eco-modulation fee implementation]
- Access: [value chain actors, public authorities]

Interview UC6. Net-Zero Defect Manufacturing

Presenter: Alessia Matruglio (Atlantis) and Joel Sepúlveda (Innovalia).

Date of Interview: May 27th, 2025.





UC Overview:

This use case aims to monitor (and reduce) the existence and the amount of pollution emissions—specifically CO₂ and methane—by integrating satellite image data and on-site industrial machines emissions and energy usage data. The objective is to contribute to the goal of net-zero emissions.

Geographical scope: not determined yet (*ask again*).

Stakeholders involved: needs to be specified.

General data aspects:

- Atlantis is working on implementing STAC (SpatioTemporal Asset Catalog).
- File size and format: not confirmed during the interview but crucial for integration with Eclipse Data Connector.

Data access: currently done through Atlantis's own internal system but could be potential integrated into the GDDS.

Data flows identified:

1. CO₂ and methane emission data sharing

- Input: satellite processed image data and metadata ([STAC](#)), on-site industrial machines emissions and energy usage data.
- Processing: combination of satellite imagery (from Atlantis) with on-site machine emission data (from Innovalia) and generation of integrated datasets for monitoring and decision support.
- Output: High-resolution images showing methane and CO₂ levels and reports quantifying and characterising energy and emission data (CO₂ and methane real-time monitoring).
- Access: results will only be shared with authorized users.

Interview UC7. Climate Investment Plan Dataspace (CIPD)

Presenter: Fedra Vanhuyse (SEI).

Date of Interview: July 2nd, 2025.

UC Overview:

This use case focuses on enabling cities to estimate the investment required to become climate neutral by 2030. In Sweden 48 different cities that represent over 50% of Swedish population that have committed to climate neutrality by 2030. This UC has developed an open-source modelling tool (a python platform - the [Viable Cities Finance Dashboard](#)) that allows Swedish cities to assess their current emissions, simulate different climate scenarios (such as adopting electric vehicle), calculate associated costs and benefits, and identify funding opportunities and financial gaps. (the impact of various climate actions). They have used publicly available data from the Swedish Environmental Protection Agency, the Swedish Statistical Agency, etc., around how much CO₂ does every city emit and what kind of assets are there in each city.

Geographical scope: 90 Swedish cities ([Viable Cities Finance Dashboard](#)) and 112 European cities participating in the EU Mission for Climate-Neutral and Smart Cities ([NetZeroCities](#)).

Stakeholders involved: national, regional and local governments and public agencies.



General data aspects: they have identified 2 challenges. The first challenge of this UC is about data availability, more specific to the level of granularity of the data. Often data is available at a national level that then need to be downscaled to the local level which leads to a lot of error margin. The second challenged identified is about the confidentiality of the data (geospatial) due to geopolitical sensitivity.

Data flows identified: the following dataflows have been identified:

1. Harmonised climate and socioeconomic data for effective climate planning

- Input: 1000 data points (indicators) around CO2 emission data, data about existing assets (number of cars, busses, etc., and how many of them are electric or biofueled), average energy consumption of the buildings) and socioeconomic data such as population over time.
- Processing: the role of the GDDS is not to run the model itself but to act as a trusted, interoperable platform that improves access to high-quality, harmonized, and securely shared data. This could enable municipalities, researchers, and public agencies to collaborate more effectively on climate planning.
- Output: [Dashboard](#) (visualization of a model of a calculation). There are 6 databases: Emissions; Assets and behaviour in the city; Climate action, mitigation potential and the cost and benefits of each measure; Co-benefits of each action; Available subsidies and grants for these climate actions.
- Access: national, regional and local governments, public agencies, and private companies.

Return to the DS: a new dataset. The stakeholders, the data contributors (governments, municipalities, private companies like google) should be required to be part of the Data Space so that the UC can benefit from these data. As a result, there's the dashboard which returns very personalised results and so there's not a return to the Data Space.

Interview UC8. Environmental hazards

Presenter: Karssenberg, D.J. (UU).

Date of Interview: July 3rd, 2025.

UC Overview:

This use case aims to provide harmonized, high-resolution environmental exposure data to support public health research and policymaking. The use case processes large-scale raster data (air pollution, temperature, noise, green space) to generate personalized and aggregated exposure maps. These are made accessible through a data catalogue, APIs, and a map viewer, enabling researchers and institutions to enrich cohort studies or conduct epidemiological analysis without directly handling raw environmental data.

Geographical scope: the data covers primarily Europe.

Stakeholders involved: research institutions, public health authorities and policymakers, infrastructure providers, general public, environmental data providers.

General data aspects: Data resolution is 100 meters and outputs can be aggregated at city, province, or country level.



Data flows identified: the following dataflows have been identified:

1. Exposure map generation and dissemination

- Input: environmental datasets primarily in raster format (air pollution, noise, temperature, green space, etc.) sourced from public and open-access sources (Copernicus, ECMWF, etc.), and population data (this will be done in their own systems as well as the processing).
- Processing: generation of exposure maps, aggregated statistical data.
- Output: generation of personalized exposure maps per variable (GeoTIFF) in different levels of detail.
- Access: via catalogue and basic map viewer. Users can download the full dataset or only a subset by region, coordinates, or address. The use can also retrieve the data via API. Future potential for paid access.

2. Data enrichment for epidemiological research

- Input: cohort data from health researchers (addresses, GPS tracks or coordinate lists).
- Processing: matching external user data (e.g., location points) with the processed exposure maps.
- Output: user coordinates with environmental exposure values. The output supports epidemiological correlation studies.
- Access: secure access depending on user role (researcher, policymaker, etc.). It won't be open to the public due to potential sensitivity of usage.

Interview UC9. Nature and ecosystem services trade-offs (NECST) assessment tool

Presenter: Alicia Martinez Varela (NTTDATA), Florian Jean Caumont (NTTDATA), Marc Gil Garriga (NTTDATA).

Date of Interview: July 18th, 2025.

UC Overview:

This use case aims to pilot the integration of the Green Deal Data Space (GDDS) with a Nature and Ecosystem Services Trade-offs (NECST) assessment tool, demonstrating how a centralized, interoperable data infrastructure can significantly enhance environmental analysis.

Geographical scope: European, national and regional level.

Stakeholders involved: two distinct company types and public data providers.

Data flows identified: the following dataflows have been identified:

1. Integration of GDDS-NECST to enhance sector-specific environmental analysis

- Input: selected datasets will be extracted from the GDDS and transformed into formats compatible with the NECST tool. This may involve harmonizing spatial and temporal resolutions, aligning classification systems, and ensuring semantic interoperability.
- Processing: selected datasets will be extracted from the GDDS and transformed into formats compatible with the NECST tool. This may involve harmonizing spatial and temporal resolutions, aligning classification systems, and ensuring semantic interoperability. The NECST tool will be configured to reflect the sector-specific context of each company type, incorporating the adapted data to model trade-offs between ecosystem services and economic activities.



- Output: The outputs of the NECST assessments will be compared across the two company types to evaluate how different sectors interact with nature, and how the GDDS can support tailored, data-driven sustainability assessment. Crucially, this step is conducted from the perspective of corporate end-users. It is designed to demonstrate how companies can use NECST to gain actionable insights that are directly relevant to their operations, investment decisions, and sustainability planning.
- Access: companies interested in the output results.

Interview UC10. Air Quality and Health Dataspace (EIRENE)

Presenter: Ian Michael Hedgecock (CNR), Francesco de Simone (CNR).

Date of Interview: July 2nd, 2025.

UC Overview:

This use case focuses on the [EIRENE](#) project, a European research infrastructure aiming to study the human exposome (the total environmental exposures individuals face from conception to death) and how these influence health results (or outcomes). The project integrates environmental, occupational, and health data to identify links between exposures (e.g., air pollution, temperature anomalies) and health events (e.g., hospitalizations), supporting research, policy-making, and potential public health interventions.

Geographical scope: European scale, with a pilot use case focused on Italy (province-level analysis). Data and studies may expand across the EU depending on data access and interoperability.

Stakeholders involved: EIRENE project team, national and local governments, EU institutions and projects, researchers, national and regional health services.

General data aspects: privacy and sensitivity of health data are key constraints.

Data flows identified: the following dataflows have been identified:

1. Environmental exposure to health outcome correlation

- Input: environmental data (air pollution, climate variables, geographic data) and health data (hospital admissions such as cardiovascular and respiratory).
- Processing: statistical models are applied to identify correlations or causality between environmental variables and hospitalisation data. Focused analysis on specific regions and periods (for example heatwaves in summer).
- Output: significant variables with statistical impact on hospitalisation. Outputs include coefficients, correlation scores, and potential causality indicators.
- Access: researchers with authorised access under EU and national privacy laws. Hospitals and local government may use outputs for planning.

2. Researchers' hypothesis testing via the Data Space

- Input: environmental data (air pollution, climate variables, geographic data) and health data (hospital admissions such as cardiovascular and respiratory).
- Processing: hypothesis-driven modelling.
- Output: case-specific insights and reports with a potential basis for scientific publications, policy decisions or public health interventions.
- Access: researchers, policymakers access the outputs in a secure manner. This enables research while preserving privacy.





Appendix B: Extraction of preliminary storylines based on the identified use case dataflows

DISCLAIMER: In the next months, these storylines will be refined and perfected with the materials collected in the trial handbooks. Please consider this as a preliminary attempt to extract storylines.

UC 1. Forest Dataspace (FDS) Cluster
As a data analyst I want to create a BPOD (open and protected) in the GDDS in order to analyse it.
As a forest owner I want to share synthetic training datasets with end users instead of raw sensitive forest inventory data in order to protect privacy.
As a forest owner I want a live monitoring of my trees in order to detect losses or fatalities early and make management decisions at a suitable time.
As a forest data manager (Data Holder) I want to access derived datasets from forest transformation simulations in order to plan future forest estates under different climate and management scenarios.
As a researcher I want to access a large forest data pool in order to develop tools and models for climate-adapted forest management and biodiversity planning.
As an environmental authority I want access to harmonised and interoperable forest data in order to ensure sustainable, well-informed decision-making across multiple stakeholders.
UC 2. Pollinator Monitoring Dataspace (PMDS)
As a farmer I want recommendations based on pollinator monitoring data in order to make informed decisions about crop treatments, buffer zones, pollinator-safe products, and hive rental.
As a farmer or apiary owner I want to use matchmaking services based on spatial data, flowering calendars, and treatment alerts in order to coordinate seasonally, optimize pollination services, and improve crop yield planning.
As a policy maker I want access to aggregated pollinator health data in order to implement agricultural policies based on evidence and that are pollinator friendly.
As a researcher I want access to harmonized datasets combining geographical, taxonomic, and soil quality data in order to develop accurate AI models for pollinator detection and to analyze trends in pollinator populations and habitat quality.
UC 3. Soil Circularity Dataspace (SCDS)
As a contractor I want to supply excavation project data, including soil batches and quality, through a standardized interface in order to manage my soil balance and decide whether to inform surplus or demand on the marketplace.
As a marketplace user I want soil batches to include open data, predictive models, and legal requirements in order to find the best matches between supply and demand.
As an environmental service I want to track soil transport records and detect discrepancies in batch data in order to identify and prevent illegal activities such as mixing contaminated soil with clean soil.
As a soil data owner, I want to control the visibility of my soil data and decide whether it is private, shared with specific users, or made public in order to protect ownership while enabling collaboration in the soil marketplace.
As a construction contractor I want to find available excavation batches that meet my soil needs through the platform in order to lower costs and support circular soil practices.
As a municipality official I want to access enriched marketplace data on available soil batches and demand opportunities in order to plan sustainable construction and infrastructure projects and reduce the need for new soil extraction.



UC 4. Construction Environment CO₂ Hub & Building Twins (BECO₂)

As a real estate owner, I want to receive harmonized CO₂ emission data from service providers in order to report Scope 3 emissions consistently and comply with EU climate policies.

As a building maintenance service provider, I want to share primary Scope 3 CO₂ emission data of my operations, with its associated metadata, in order to demonstrate transparency and support clients' sustainability reporting aligned with evolving regulatory frameworks and sustainability goals.

As a public administration I want access to real estate primary Scope 3 CO₂ emission reports and carbon budget calculations in order to monitor compliance and guide policy decisions.

UC 5. Textile Cluster in GDDS

As a textile sorter, I want to access textile product information from the DPP, including fiber composition and product type data in order to improve decisions on end-of-life management such as reuse, recycling, and repair.

As a DPP provider I want to pull value chain data such as recycled content and certification data onto the DPP in order to decrease reporting burden and costs.

As a Producer Responsibility Organization (PRO) I want to harmonize data from post-consumer textile management activities and to report data to the national authorities that is harmonized with other PROs.

UC 6. Net-Zero Defect Manufacturing

As a data scientist (working at SATLANTIS) I want to process satellite imagery for methane plume detection and emission rate estimation in order to provide validated results that can be published in the GDDS for transparency and open access.

As a satellite data provider, I want to analyze satellite imagery for methane plume detection and emission rate estimation in order to enable integration with industrial methane emission datasets for accurate monitoring.

As an industrial operator I want to combine on-site methane and CO₂ emission and energy usage data with satellite imagery in order to monitor and reduce my facility's CO₂ and methane emissions.

As a policymaker I want real-time reports quantifying CO₂ and methane emissions in order to support decision-making and progress toward net-zero goals in compliance with international climate agreements.

As an end-user I want to receive methane plume datasets with secure access in order to integrate this information into my own systems and support climate action strategies ensuring compliance with international climate agreements.

UC 7. Climate Investment Plan Dataspace (CIPD)

As a municipality I want to access harmonized CO₂ emission data, data about existing assets (number of cars, busses, etc., and how many of them are electric or biofueled), and socioeconomic datasets in order to simulate climate scenarios and estimate the investments needed for climate neutrality.

As a researcher I want to access the GDDS through a dashboard to analyze high-quality, securely shared indicators in order to support evidence-based climate planning and scenario modeling.

As a public agency (or a private company like google) I want a trusted platform (the GDDS) to contribute with aggregated data on emissions and assets to be able to access subsidies and grants for climate actions.

UC 8. Environmental hazards

As a health researcher I want to enrich my cohort data with environmental exposure data in order to conduct epidemiological studies linking health outcomes with environmental factors.

As a public health authority, I want access to aggregated exposure maps of air pollution, noise, temperature, and green space in order to design evidence-based policies to improve population



health.
As a policymaker I want to retrieve high-resolution exposome raster maps via APIs and catalogues in order to evaluate health risks and guide spatial planning decisions.
UC 9. Nature and ecosystem services trade-offs (NECST) assessment tool
As a data provider I want to contribute harmonized datasets to the GDDS in order to enable their integration into the NECST tool for consistent and comparable analysis.
As a corporate end-user I want to access NECST assessments derived from GDDS data in order to gain actionable insights for my operations, investment decisions, and sustainability planning.
UC 10. Air Quality and Health Dataspace (EIRENE)
As a researcher I want to access harmonized environmental and health datasets from the GDDS in a secure way in order to test hypotheses about correlations between exposures and health outcomes.
As a public health authority, I want statistical outputs on correlations between air pollution, climate anomalies, and hospital admissions in order to plan interventions and allocate resources effectively.
As a policymaker I want access to the GDDS to retrieve case-specific insights from exposome data analysis in order to design evidence-based policies that protect population health (make better-informed decisions)
As a hospital administrator I want access to regional health–environment correlation reports in order to anticipate healthcare demand during events such as heatwaves.